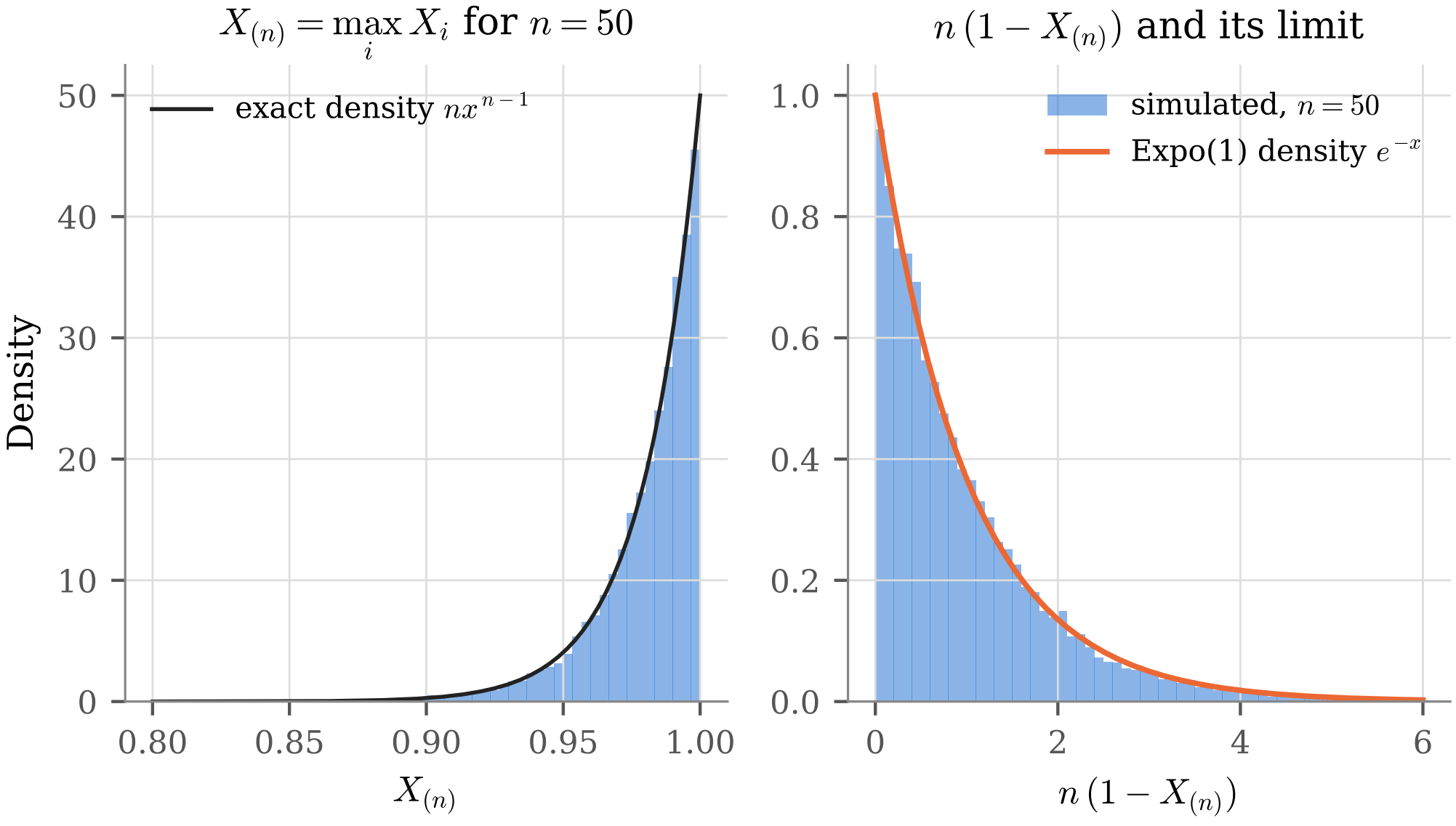
$X_{(n)} = \max_i X_i$ for $n = 50$
exact density nx^{n-1}
Density
50
40
30
20
10
0
0.80
0.85
0.90
0.95
1.00
$X_{(n)}$
$n\,(1 - X_{(n)})$ and its limit
simulated, $n = 50$
Expo(1) density e^{-x}
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
$n\,(1 - X_{(n)})$